Nova AI -- Engineering Practices

Development Workflow

We follow a trunk-based development model with short-lived feature branches (max 3 days). Every branch must pass a CI pipeline that includes type checking (mypy), linting (ruff), unit tests (pytest, 80% coverage minimum), and integration tests before merging to main.

Code Review Standards

- Every PR requires at least one approval from a senior engineer on the team.
- Reviews must be completed within 24 hours during business days.
- PRs larger than 400 lines require a design doc and sync review meeting.
- All feedback must be constructive and reference specific line numbers.
- Automated checks must pass before human review begins.

Tech Stack

Backend:

Python 3.12, FastAPI, LangGraph, SQLAlchemy (async), PostgreSQL (Supabase), Redis, Pinecone

Frontend:

React 19, TypeScript, Tailwind CSS, Vite, React Router v7

AI/ML:

OpenRouter (primary LLM provider), Groq (fallback), LangChain, sentence-transformers, Whisper

Infrastructure:

Docker, GitHub Actions (CI/CD), Vercel (frontend), Render (backend), Supabase (DB + auth)

Deployment Process

Deployments happen automatically on merge to main after a 10-minute smoke test window. Production deploys use a blue-green strategy with zero downtime. Rollbacks are automated if error rates exceed 1% within 5 minutes of deploy.

Monitoring & Observability

We use structured logging (JSON) shipped to a central log aggregator. Key metrics (p95 latency, error rate, token usage, cost-per-interview) are tracked via a custom dashboard with PagerDuty alerts on thresholds. Every service exposes a /health endpoint with dependency status.